§ 12 Perturbation 2

$H = H_0 + V(t)$

1.

① 瞬时本征态

$H(t)|n(t)\rangle = E_n(t)|n(t)\rangle$, $|\psi(t)\rangle = \sum_n C_n(t)|n(t)\rangle$

Schrödinger Eq.

$i\hbar \partial_t |\psi(t)\rangle = \sum_n i\hbar \dot{C}_n |n\rangle + C_n i\hbar \partial_t |n\rangle = \sum_n C_n E_n |n\rangle$

$\Rightarrow i\hbar \dot{C}_n = E_n C_n - \sum_m i\hbar C_m \langle n|\partial_t|m\rangle$

② 取 H_0 的基矢

$H_0|n\rangle = E_n|n\rangle$

$|\psi(t)\rangle = \sum_n C_n(t) e^{-\frac{iE_n t}{\hbar}} |n\rangle$

$i\hbar \partial_t |\psi(t)\rangle = \sum_n i\hbar \dot{C}_n(t) e^{-\frac{iE_n t}{\hbar}} |n\rangle + E_n C_n(t) e^{-\frac{iE_n t}{\hbar}} |n\rangle$

$= H|\psi(t)\rangle = \sum_n C_n(t)(E_n + V) e^{-\frac{iE_n t}{\hbar}} |n\rangle$

$\Rightarrow i\hbar \dot{C}_n = \sum_m \langle n|V|m\rangle e^{\frac{i(E_n - E_m)t}{\hbar}} C_m$

$= \sum_m V_{nm} e^{i\omega_{nm} t} C_m$ $\quad \omega_{nm} = \frac{E_n - E_m}{\hbar}$

也可以从概率作用绘景出发.

展开：$C_n(t) = C_n(0) + \sum_m \frac{1}{i\hbar}\int_0^t dt' V_{nm}^I(t') C_m(t')$

$= C_n^0 + C_n^1 + C_n^2 + \cdots$

$C_n^1(t) = \sum_m \frac{1}{i\hbar}\int_0^t dt' V_{nm}^I(t') C_m^0$

$C_n^2(t) = \sum_{m,k} \left(\frac{1}{i\hbar}\right)^2 \int_0^t dt' V_{nm}^I(t') \int_0^{t'} dt'' V_{mk}^I(t'') C_k^0$

3. 一阶微扰

$$V(t) \sim \begin{cases} e^{-t/\tau} \\ e^{i\Omega t} \\ e^{t/\tau} \end{cases}$$

(1) $|\psi^{(0)}(0)\rangle = |1\rangle$, $V(t) = e^{-t/\tau}$

$$C_{n0}(0) = \delta_{n1}, \quad V_{nm}(t) = V_{nm}(0)\, e^{-t/\tau}$$

$n=1$:

$$C_1(t) = 1 + \frac{1}{i\hbar}\int_0^t dt'\, V_{11}(0)\, e^{-t/\tau} \quad \text{无 } V_{11} \text{ 吸收到 } H_0 \text{ 中}$$

$n\neq 1$:

$$C_n(t) = 0 + \frac{1}{i\hbar}\int_0^t dt'\, V_{n1}\, e^{-t/\tau}\, e^{i\omega_{n1}t}$$

$$= \frac{V_{n1}}{i\hbar}\cdot\frac{1}{i\omega_{n1}-\frac{1}{\tau}}\, e^{\left(i\omega_{n1}-\frac{1}{\tau}\right)t}$$

$$|C_n(t)|^2 = \frac{V_{n1}^2}{\hbar^2}$$

(2) $|\psi^{(0)}\rangle = |1\rangle$, $V(t) \sim e^{i\Omega t}$

Fermi Golden Rule

$$\int = \frac{t}{\hbar^2}|V_{n1}|^2\, \delta\left[t(\omega_{n1}-\Omega)\right]$$

$$\langle V_{n1}(\Omega) V_{n1}^*(\Omega')\rangle = S_{n1}(\Omega)\,\delta(\Omega-\Omega')$$

$$V(t) = \int d\Omega\, V(\Omega)\, e^{-i\Omega t}$$

$$C_n(t) = -\frac{1}{\hbar}\int d\Omega\, V(\Omega)\, e^{-i\Omega t}\, \frac{e^{i(\omega_{n1}-\Omega)t}-1}{\omega_{n1}-\Omega}$$

$$|C_n(t)|^2 = \frac{1}{\hbar^2}\int d\Omega\, V(\Omega)\, e^{-i\Omega t}\, \frac{e^{i(\omega_{n1}-\Omega)t}-1}{\omega_{n1}-\Omega}\int d\Omega'\, V^*(\Omega')\, e^{i\Omega' t}\, \frac{e^{i(\omega_{n1}-\Omega')t}-1}{\omega_{n1}-\Omega'}$$

Let $S_{n1}(\Omega)\,\delta(\Omega-\Omega') = \int dt'\, V^*(\Omega')\, V(\Omega)\, e^{i(\Omega-\Omega')t'}$

ex) $|\psi^2(-\infty)\rangle = |1\rangle$, $V(t) \sim e^{\Gamma t}$

4.

시간 E - t uncertainty.

$\Delta t \cdot \Delta E \sim \hbar$

① Resonate State.

②

(2) Stimulated Radiation

(3)